Name

Class

R. No.

- Mathematics
- IIT JEE

- Test 4(July 16 , 2010)
- Relations

- Maximum Marks 100
- Time 60 minutes

Instructions

1. **The test contains nine subjective questions**
2. **The marking scheme will include marks for partially correct answers also.**

Q1: Let R and R' be two relations in a set A. Prove the following:

i) R and R' both are symmetric $\Rightarrow$ $R \cup R'$, $R \cap R'$ both are symmetric .

ii) R and R' both are transitive $\Rightarrow$ $R \cap R'$ is transitive but $R \cup R'$ needs not be transitive.

iii) R and R' both are reflexive $\Rightarrow$ $R \cup R'$, $R \cap R'$ both are reflexive .

iv) R is symmetric $\Rightarrow$ R^{-1} is symmetric and $R^{-1} = R$ 20 Marks

Q2: Show that the relation R in the set of real numbers defined by $R = \left\{(a, b) : \sqrt{a} + 3 \leq b\right\}$ is neither reflexive, nor symmetric , nor transitive.
10 Marks

Q3: Let R be a relation in the set N. Let$R = \{(x, y) \in N \times N : x + 2y = 10\}$. Find the domain and range of R and R^{-1} .
4 Marks

Q4: A relation R is defined on the set $N \times N$, where N is the set of natural numbers, by setting $(a, b)R(c, d)$ if and only if $a + d = b + c$. Show that the relation is an equivalence relation . Describe the equivalence class of $(2,3) \in N \times N$. 8+4 Marks

Q5: If R and R' be two equivalence relations on a set A. Prove that

i) $R \cap R'$ is an equivalence relation in A.

ii) $R \cup R'$ needs not be an equivalence relation in A.

14 Marks

Q6: Let $A = \{1, 2, 3 \ldots \ldots \ldots \ldots 14, 15\}$. Let R be the equivalence relation on A defined by congruence modulo 4.

i) Find the equivalence classes determined by R.

ii) Find a system B of equivalence class representatives which are multiples of 3. 3+2 Marks

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Q7 : Each of the following expressions defines a relation on $\mathbb{R}$

i) $x^2 - 4y^2 \geq 9$

ii) $x^2 + 4y^2 \leq 16$

Sketch each relation in the plane $\mathbb{R}^2$. 10 Marks

Q8: Give an example of a relation , Which is

i) Symmetric but neither reflexive nor transitive.
ii) Transitive but neither reflexive nor symmetric.
iii) Reflexive and symmetric but not transitive.
iv) Reflexive and transitive but not symmetric.
v) Symmetric and transitive but not reflexive.
vi) Both symmetric and antisymmetric
vii) Neither symmetric nor antisymmetric

14 Marks

Q9 Show that the relation R in the set A of points in a plane given by R={(P,Q): distance of the point P from the origin is same as the distance of the point Q from the origin} , is an equivalence relation. Further, show that the set of all points related to a point P(not the origin) is the circle passing through P with origin as the centre. 6+5 Marks